Bocconi

CORPORATE FINANCE 30017

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Markets and Information

Università Bocconi
MILANO

Agenda

MARKET EFFICIENCY

EVENT STUDY METHOD

MARKET EFFICIENCY

Information and Prices

$$P_{i,0} = \sum_{t=1}^{\infty} \frac{\boldsymbol{E}[c_{i,t}]}{(1+\delta_i)^t}$$

- Some **price movements** are expected, driven by realized-as-expected cash flows and/or capital gain

- But empirically, most price movements are **unexpected**, driven by **new information** about:
  - Realized cash flows
  - Expected future cash flows
  - Perceived risk of future cash flows
- News may affect individual assets or many simultaneously (market vs. idiosyncratic shocks)

What Information priced in when?

Weak Market Efficiency:
Current market prices reflect past public and private information (incl. past prices)

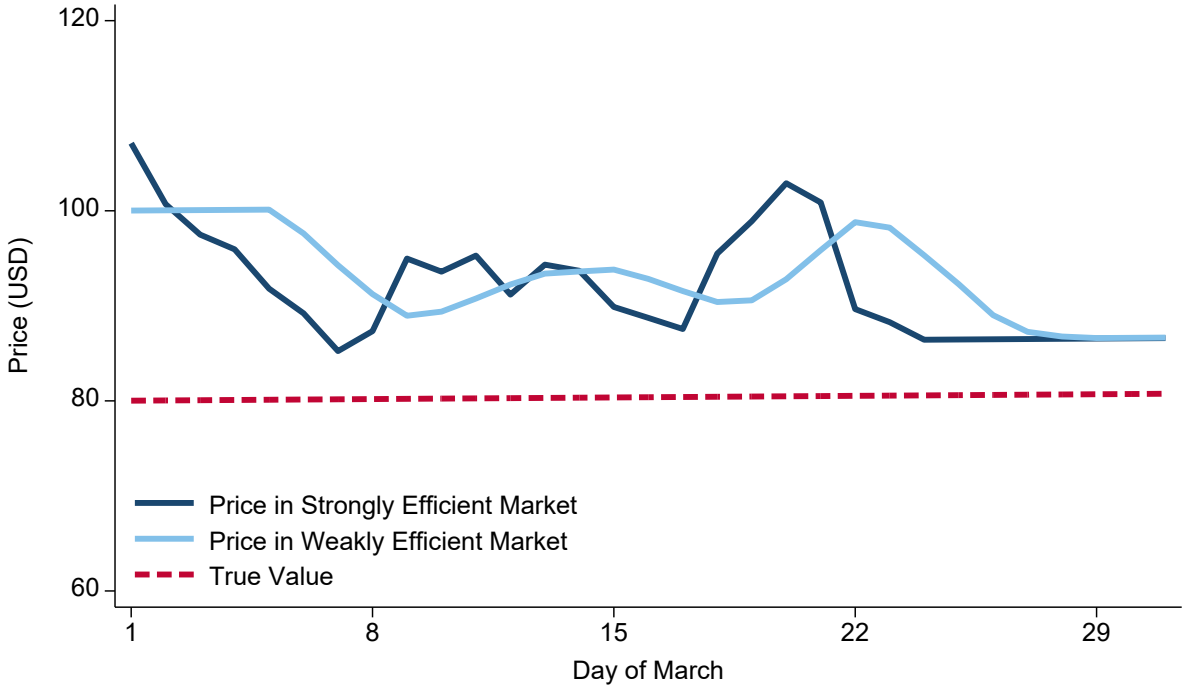
Semi-Strong Market Efficiency:
Curren market prices reflect past public and private as well as current public information

Strong Market Efficiency:
Current market prices reflect all information (current, past, public, private)

Market Efficiency Definitions: Caveats

- If some info not priced in *now*, how long until it will? Millisecond? Minutes? Weeks?
- Infinitely many shades between “private” and “public” information
  - *Is prediction of hedge fund’s market data-based pricing model private info?*
  - *Is post on wallstreetbets public info?*
- Information contains noise; does not necessarily point in direction of true value

Simulation Example



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— Less efficient financial markets tend to be more “stale” (less price movements)

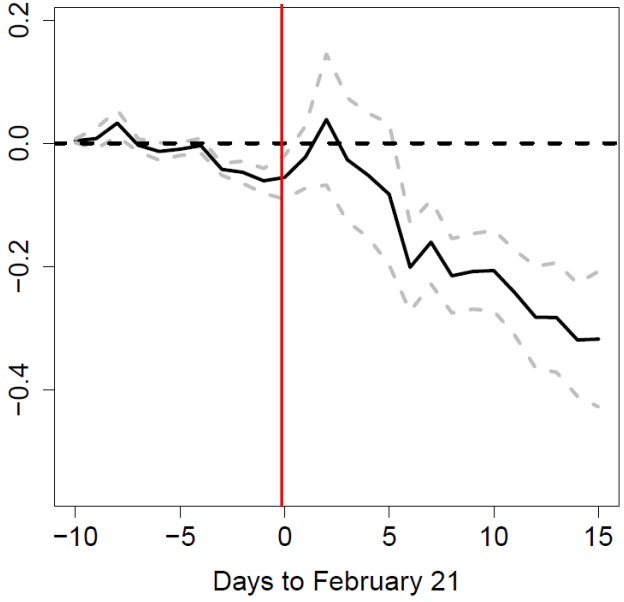
EVENT STUDY METHOD

Fundamental Idea

- Notion of (informationally) efficient markets leads to powerful **research methodology**:
  - If security prices reflect (nothing but) available infos, then price changes must reflect new infos!
  - That is, examining price changes in a time-window around the event allows us to:
    - **measure value of an event**
      - Classical example: *Announcement of M&A transaction on the bidder and target firm*
    - **measure market efficiency** (i.e., speed of information pricing)

The first EU Covid Case and Italian Stocks

Cumulative returns of Italian stocks around first Italian Covid-19 case



Complication

- Absent strong market efficiency, pricing info takes time and is uneven across info types
  - That is, **any movement** may reflect **mix** of past information.
    - *On any day, stock prices respond to wide range of news from all over the economy (and abroad)…*
  - Isolating the part of a stock price movement that is attributable to a specific event is not a trivial

- Need a **counterfactual**: *"What <u>would</u> have happened if that <u>particular event had not occurred</u>?"*

Abnormal Returns

- Proxy counterfactual return via a "**benchmark model**" predicting return $\hat{r}_{i,t}$
- Difference between that proxy and actual return is referred to as **abnormal return**:

$$AR_{i,t} \equiv r_{i,t} - \hat{r}_{i,t}$$

- Several methodologies ("benchmark models") around to estimate this proxy
  - Simple approach: proxy return = market return
  - Refined approach: proxy return = expected return prescribed by asset pricing models
    - CAPM
    - Multifactor generalizations such as the Fama-French three-factor or five-factor model
  - Another approach: proxy return = average return of stocks matched according to criteria such as firm size, beta, recent performance, or ratio of price to book value per share.

CAPM Benchmark

CAPM predicts $E[r_{i,t}] - r_{f,t} = \beta_i(E[r_{m,t}] - r_{f,t}$. This suggests to construct benchmark return as

$$\hat{r}_{i,t} = E[r_{i,t}] = r_{f,t} + \beta_i(r_{m,t} - r_{f,t})$$

— I.e, compute abnormal returns using estimate of β_i together with observable data on $r_{m,t}, r_{f,t}$:

$$AR_{i,t} = r_{i,t} - r_{f,t} - \beta_i(r_{m,t} - r_{f,t})$$

- CAPM-benchmarked abnormal returns will capture idiosyncratic information only (why?)
- If we allow for <u>a non-zero</u> abnormal return:

$$AR_{i,t} = r_{i,t} - r_{f,t} - \beta_i(r_{m,t} - r_{f,t}) - \alpha_i$$

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- Simplification with the assumption that $r_{f,t} = 0$:
  - Plausible approximation for event studies with **short** event windows

$$AR_{i,t} = r_{i,t} - \beta_i r_{m,t}$$

Running Simple Event Study

— Estimate univariate regression:

$$AR_{i,t} = \gamma_0 + \gamma_1 D_t + e_t$$

- D_t is a binary indicating the time period(s) of the events
- γ_1 directly measures the abnormal return associated with events encoded by D_t

— You may encounter various variants of this design in applied research

- E.g., run AR computation and event study in same regression:

$$r_{i,t} = r_{f,t} + \beta(r_{m,t} - r_{f,t}) + \gamma_0 + \gamma_1 D_t + e_t$$

Estimating Dynamic Event Study

— Estimate multivariate regression:

$$AR_{i,t} = \gamma_0 + \sum_{h=-H}^{H} \gamma_{1,h} D_{t-h} + e_t$$

— This estimates average **profile of abnormal returns** over window $[t-H, t+H]$ around events

- Coefficients $\{\gamma_{-H}, \dots \gamma_{-1}\}$ capture *leakage*
  - Investors with private information may trade before announcements
  - Changing prices days or weeks before

Cumulative Abnormal Return

— For some events, market may react at $t+1$, for other at $t+3$, for others….

- Dynamic event study of abnormal return may fail to detect significant effect if pricing lags vary
- Solution: **Cumulative Abnormal Returns** (expressed in percentage points):

$$CAR_i(t_1, t_2) = \sum_{t \in [t_1, t_2]} AR_{i,t}$$

— After dynamic event study regression, compute (trajectory of) average CAR:

$$\overline{CAR_i}\,(t-H, t+H) = \sum_{h=-H}^{H} \hat{\gamma}_{1,h}$$

- Inference (SE, confidence interval) can be obtained from underlying regression
- Can also run regression directly on CARs: $CAR_i(t-H, t+H) = \mu_0 + \mu_1 D_t + e_t$

Example

- The CEO of firm A unexpectedly announces to step down
- How do shareholders value the dismissal notice of the CEO?
  - Assume $\beta_A = 1.5$ and $r_{f,t} = 0\ \forall t$

days to event	return of A	market return	AR	CAR
-5	1.00%	1.00%	-0.50%	-0.50%
-4	-2.00%	1.00%	-3.50%	-4.00%
-3	1.00%	-1.00%	2.50%	-1.50%
-2	2.00%	-2.00%	5.00%	3.50%
-1	2.00%	1.00%	0.50%	4.00%
0	11.00%	-1.00%	12.50%	16.50%
1	2.00%	1.00%	0.50%	17.00%
2	-1.00%	-2.00%	2.00%	19.00%
3	1.00%	1.00%	-0.50%	18.50%
4	2.00%	2.00%	-1.00%	17.50%
5	-1.00%	-1.00%	0.50%	18.00%

$$AR_{t=0} = 0.11 - 1.5 \times (-0.01) = 0.125$$

- *Shareholders seem to appreciate the CEO's dismissal notice!*

Examples of Use outside Research

- SEC uses event studies to measure illicit gains captured by traders who may have violated insider trading or other securities laws
- Event studies are also used in fraud cases, where the courts must assess damages caused by a fraudulent activity.

How does the Market react to Data Breaches?

From *Blackboard*: CRSPDataBreach.txt

Company	Ticker	Date	Records
eBay	EBAY	06/21/14	145,000,000
Equifax	EFX	09/07/17	143,000,000
Heartland	HPY	01/20/09	130,000,000
Target	TGT	12/13/13	110,000,000
Sony	SNE	04/20/11	102,000,000
JP Morgan Chase	JPM	10/02/14	83,000,000
Anthem	ANTM	02/05/15	80,000,000
Epsilon	ADS	04/01/11	60,000,000
Home Depot	HD	09/02/14	56,000,000
TJ Maxx	TJX	02/21/07	46,000,000
Adobe	ADBE	10/03/13	38,000,000

How does the Market react to Data Breaches?

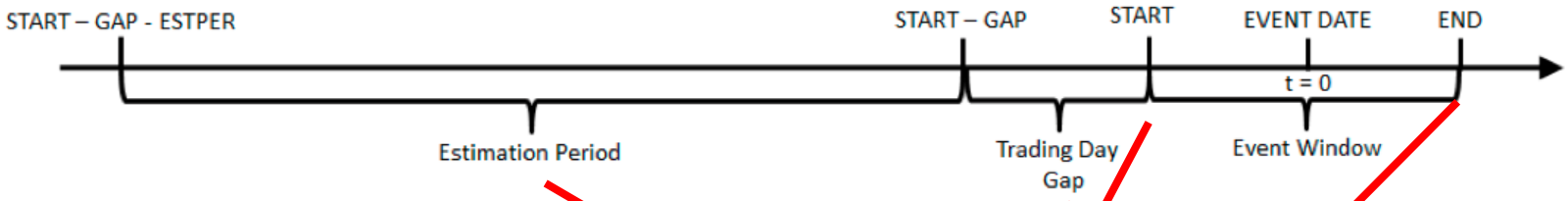
Conduct event study for the US stock market conveniently via WRDS

(WRDS also takes care of various technical bells and whistles)

Log into WRDS, navigate to “Event Study by WRDS”:

1. Select input file date format: YYYYMMDD - ticker, upload file `CRSPDataBreach.txt`
2. Select risk model (aka. benchmark model): Market model (aka. CAPM)
3. Calibration estimation: Leave at default (or see next slide)
4. Choose query variables: [all]
5. Select output format: your choice
6. *Run!*

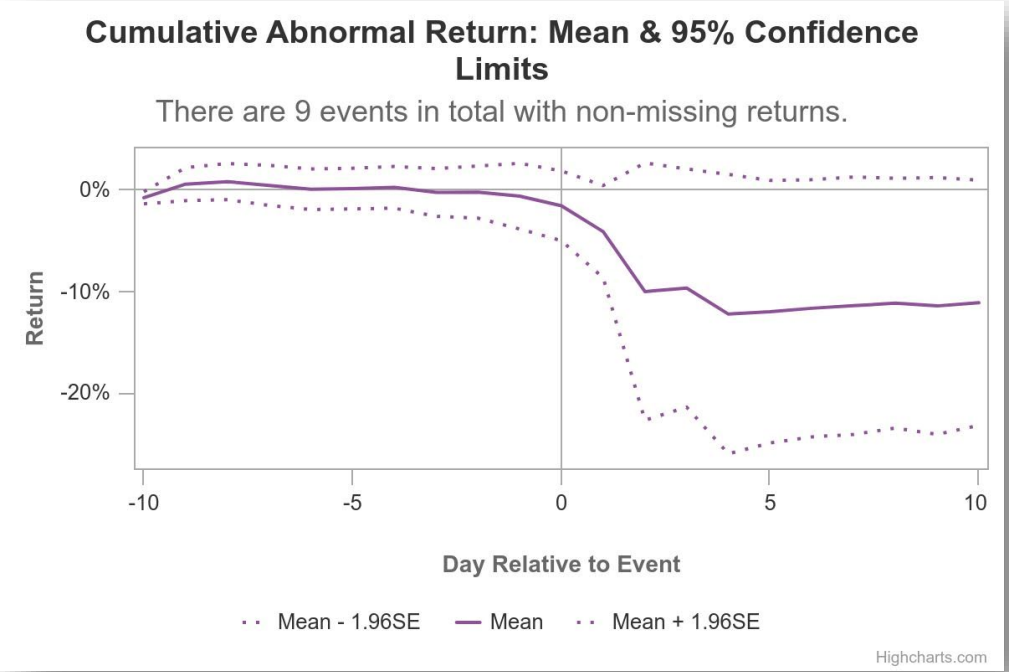
WRDS Event Study: Calibrate estimation



Step 3: Estimation parameters:

Estimation Window	100	days
Minimum Number of Valid Returns	70	observations
Gap	50	days
Event Window Start	-10	days
Event Window End	10	days

Results



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What do we conclude? Also comment on market efficiency.

Title

— Text